

Design of a Parallel Bitonic Sorter

CPEN 313 – Term Project

Due Date: May 15, 2023

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Description

Sorting algorithms are computationally expensive routines frequently executed on modern computers and embedded systems. Implementing sorting algorithms on dedicated hardware can contribute significantly to the overall execution time of the processes and applications embodying them. However, such algorithms are known to suffer from a trade off between convergence time and computational complexity. Consequently, this causes performance degradation i.e. bottleneck, when implemented on dedicated hardware with limited resources.

In this respect, a parallel hardware architecture for a Bitonic sorting network is proposed, on field programmable gate array (FPGA).

Implementation

Bitonic sorting algorithm is a comparison-based sorting algorithm that can sort a sequence of n elements in $O(\log^2 n)$ time using $O(n \log^2 n)$ comparisons. The algorithm is named after the "bitonic sequence," which is a sequence that can be divided into two parts, one of which is strictly increasing and the other of which is strictly decreasing.

The bitonic sorting algorithm works by recursively dividing the input sequence into two bitonic sequences, sorting them in a specific way, and then merging them back into a single sorted sequence.

The bitonic sorting algorithm has two phases: the sorting phase and the merging phase. In the sorting phase, the input sequence is divided into bitonic sequences. initially, the input sequence is divided into several bitonic sequences of size two, In the merging phase, each two sorted bitonic sequences are merged back into a single sorted sequence. Then, The two steps will be repeated for size of four, then size of 8, 16, etc. until we end up with one bitonic sequence, with half of it is increasing and the second is decreasing. Finally, The last merging step is performed in order to get the completely sorted data.

The bitonic sorting algorithm is often used in parallel processing applications because of its highly parallelizable nature, which makes it ideal for use in high-performance computing systems.

Project Submission

You are asked to submit a SystemVerilog for the whole design. you have the option to do the top level only in either SystemVerilog or schematics.

Your project report should include the following:

- Cover page:
 - Project Name
 - Names of students
 - University logo

- University name
- Submitted to
- Day of submission
- Objective:
 - What is the objective of the project?
 - what are you trying to do?
- Introduction: What is embedded design? Discuss some history behind it. ALoS, discuss sorting algorithms and their applications
- Description of Project:
 - Schematic: Draw each SystemVerilog as a black box and show the interconnection among all the blocks.
 - Description of each part: Clearly describe each part of your design. you may use figures and flowchart to help clarify your description.
- Alternative Solution:
 - What are some alternative solutions?
 - Why did you not choose that? For example, you may compare different implementations of the Bitonic implementations (e.g. serial vs. parallel comparisons), or compare different sorting algorithms with Bitonic such as odd-even vs. Bitonic.
- Simulation:
 - For the SystemVerilog, include several waveform figures showing several random data sets and their final sorted outputs.
- Conclusion

You are allowed to work in a group of three. However, each student will present his/her project individually. There is a major score on the presentation of the design. This score could vary from one student to another within a group depending on the performance of each student in answering questions.

The grade will be divided as follows; 30% workability, 40% report, and 30% presentation.